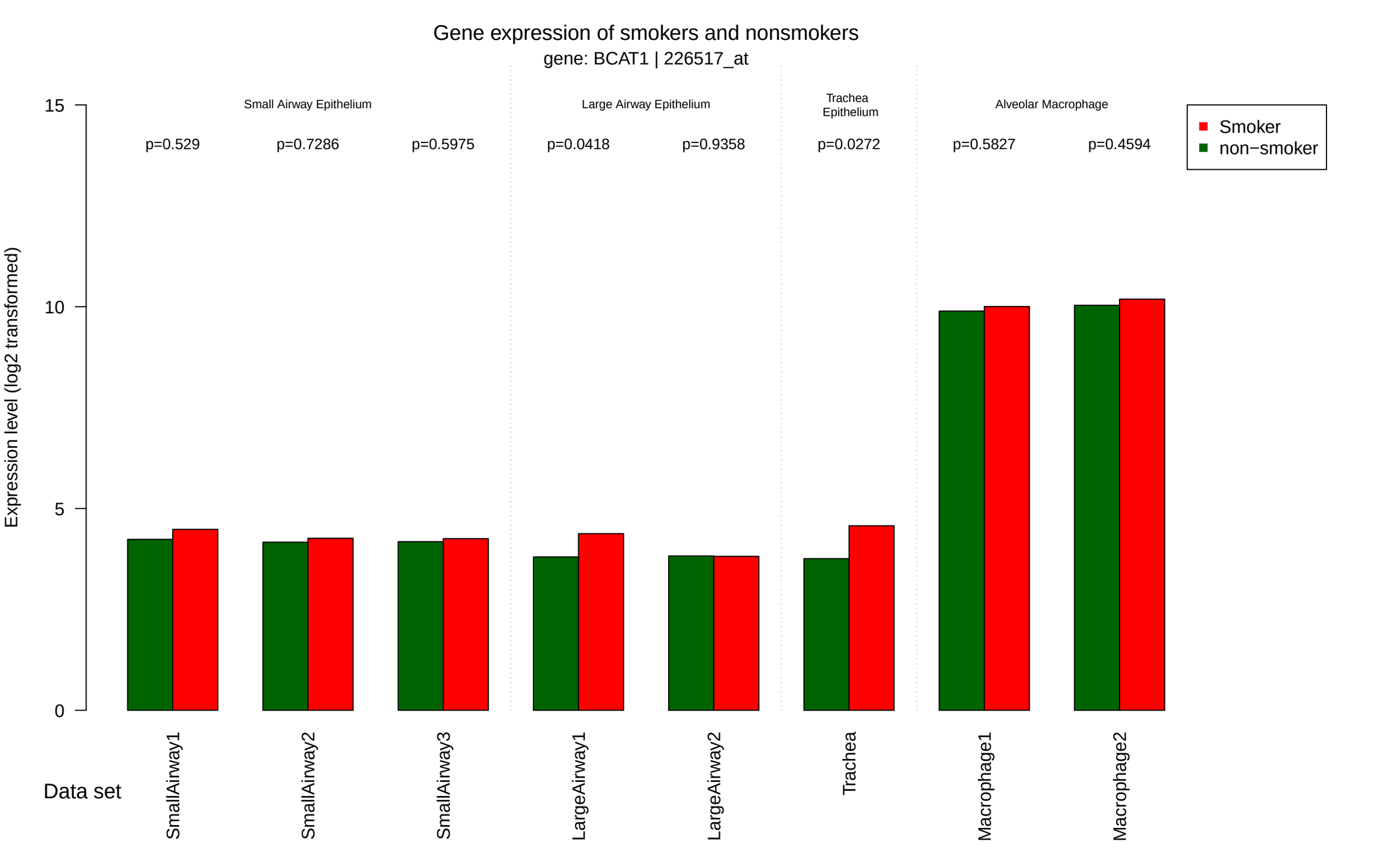
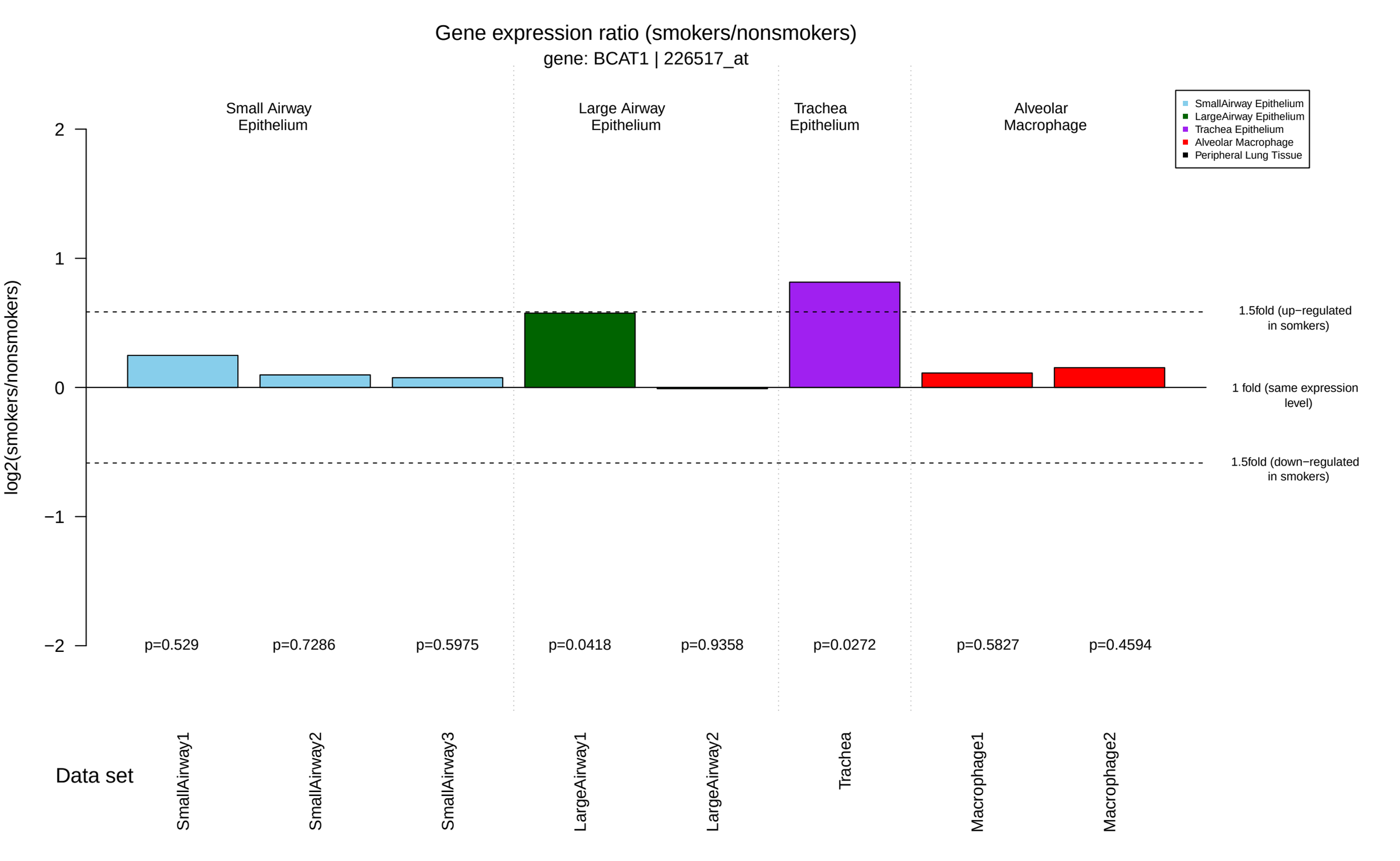
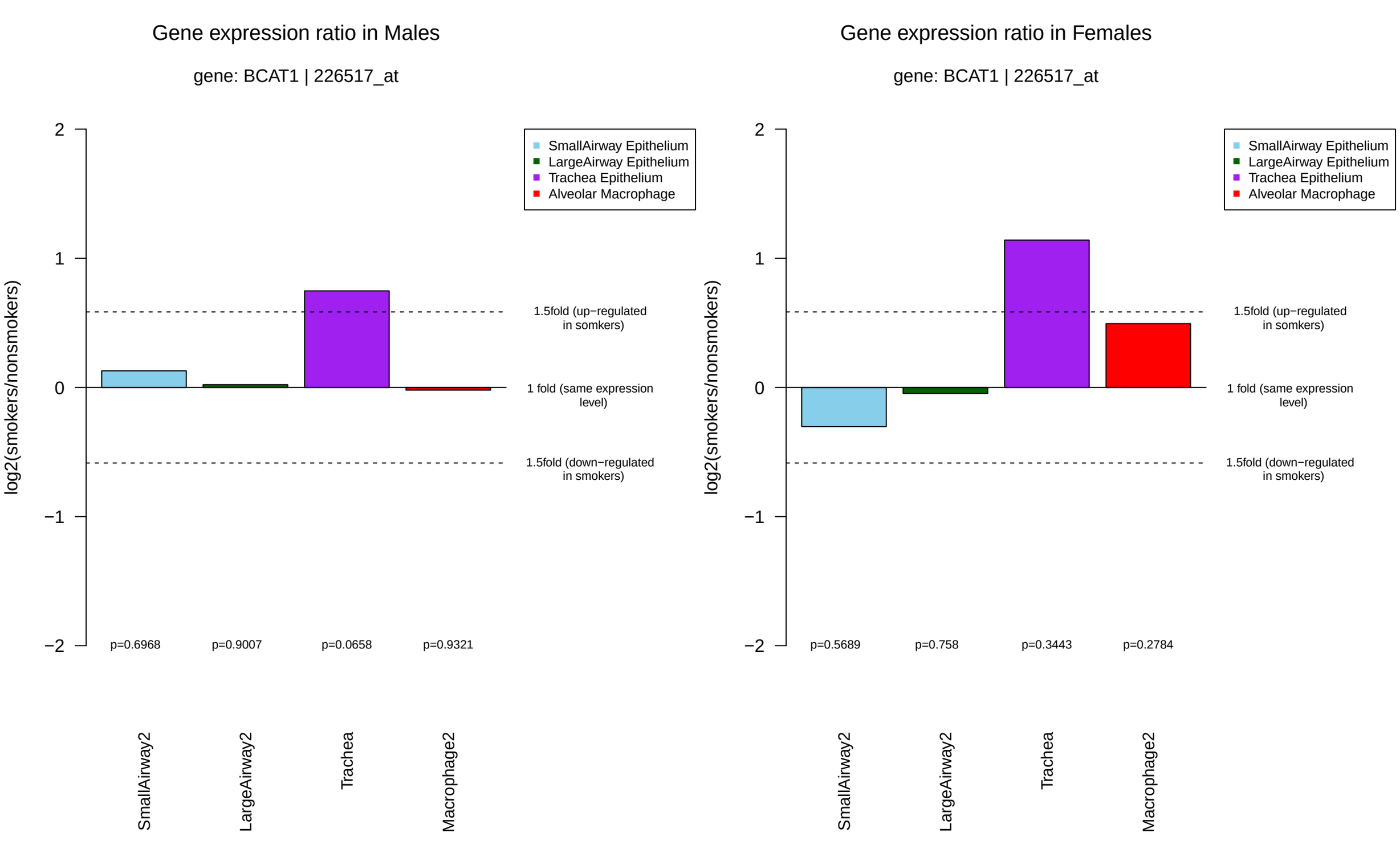
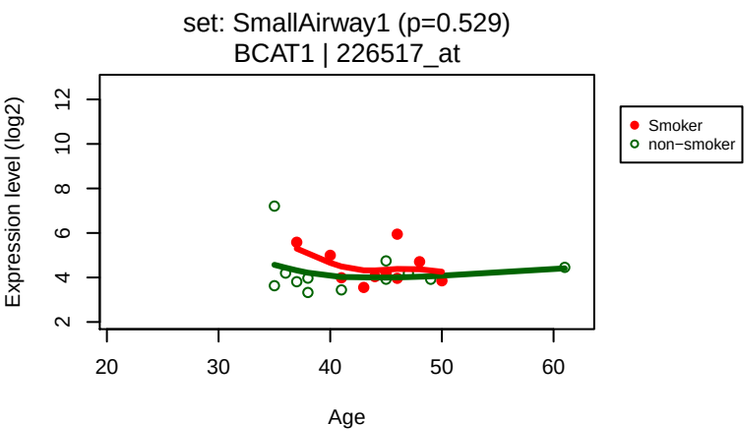
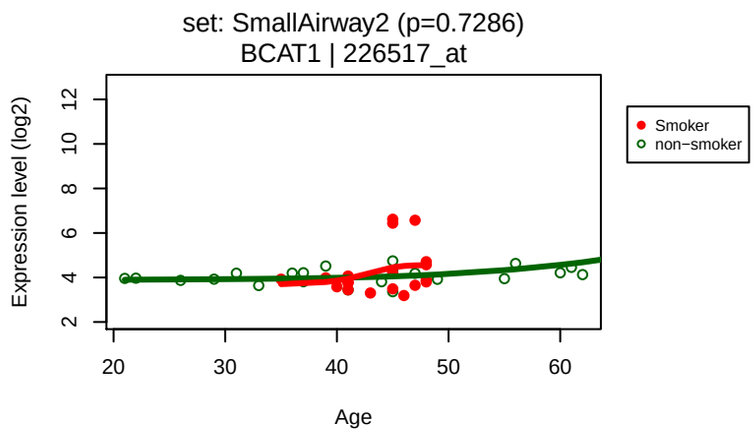
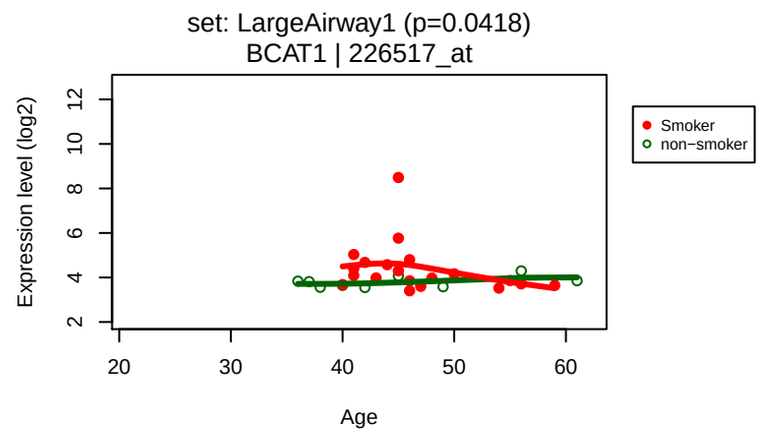

Gene expression of smokers and nonsmokers
gene: BCAT1 | 226517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.529
p=0.7286
p=0.5975
p=0.0418
p=0.9358
p=0.0272
p=0.5827
p=0.4594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

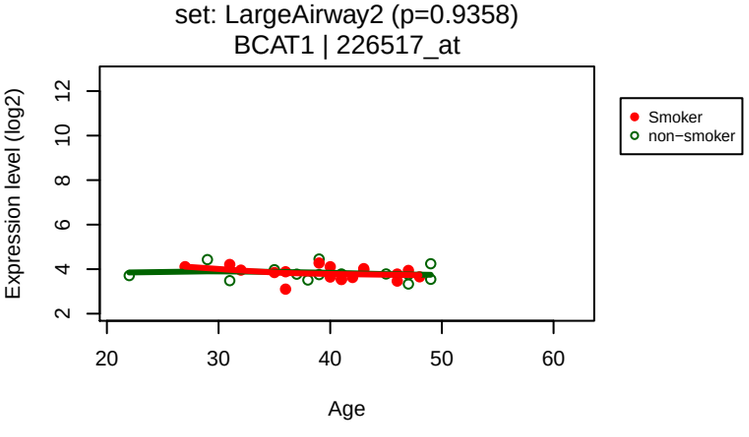

Gene expression ratio (smokers/nonsmokers)
gene: BCAT1 | 226517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.529
p=0.7286
p=0.5975
p=0.0418
p=0.9358
p=0.0272
p=0.5827
p=0.4594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

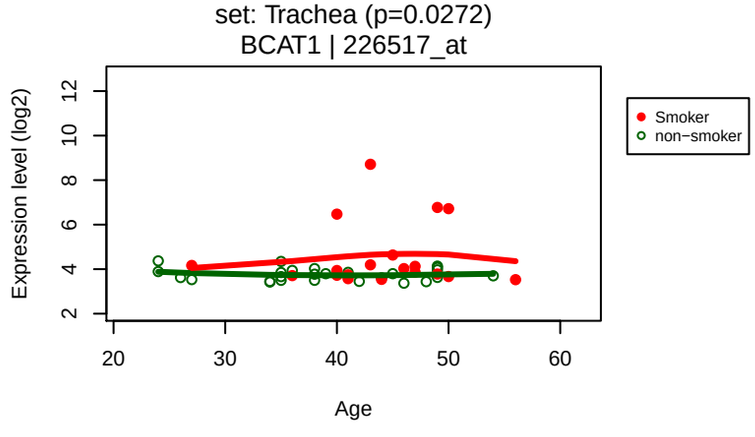

Gene expression ratio in Males
gene: BCAT1 | 226517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6968
p=0.9007
p=0.0658
p=0.9321
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BCAT1 | 226517_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5689
p=0.758
p=0.3443
p=0.2784
SmallAirway2
LargeAirway2
Trachea
Macrophage2

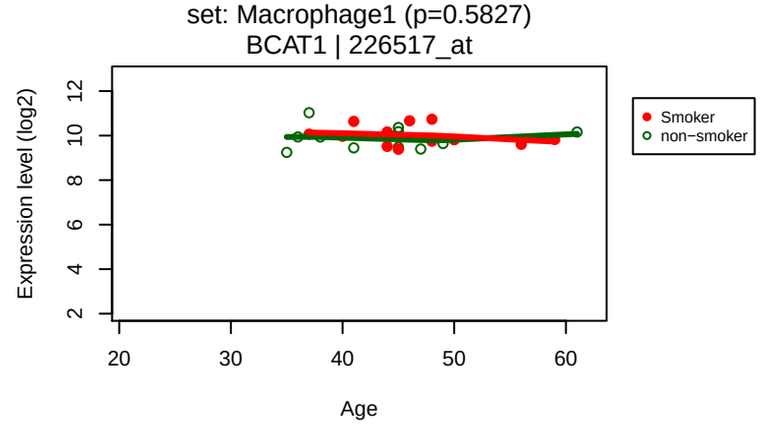

set: SmallAirway1 (p=0.529)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker

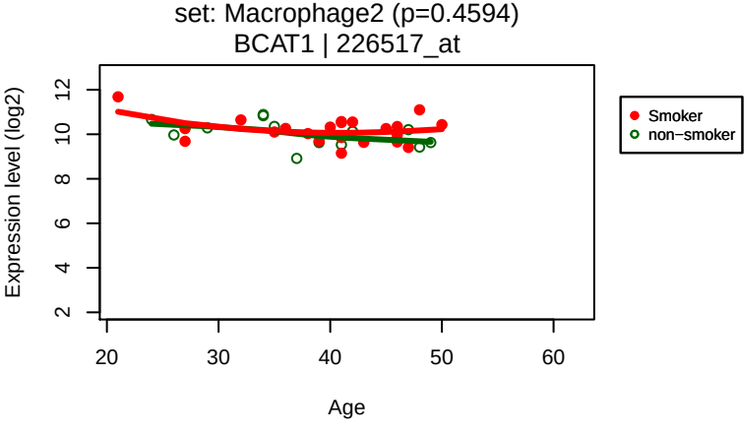

set: SmallAirway2 (p=0.7286)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker

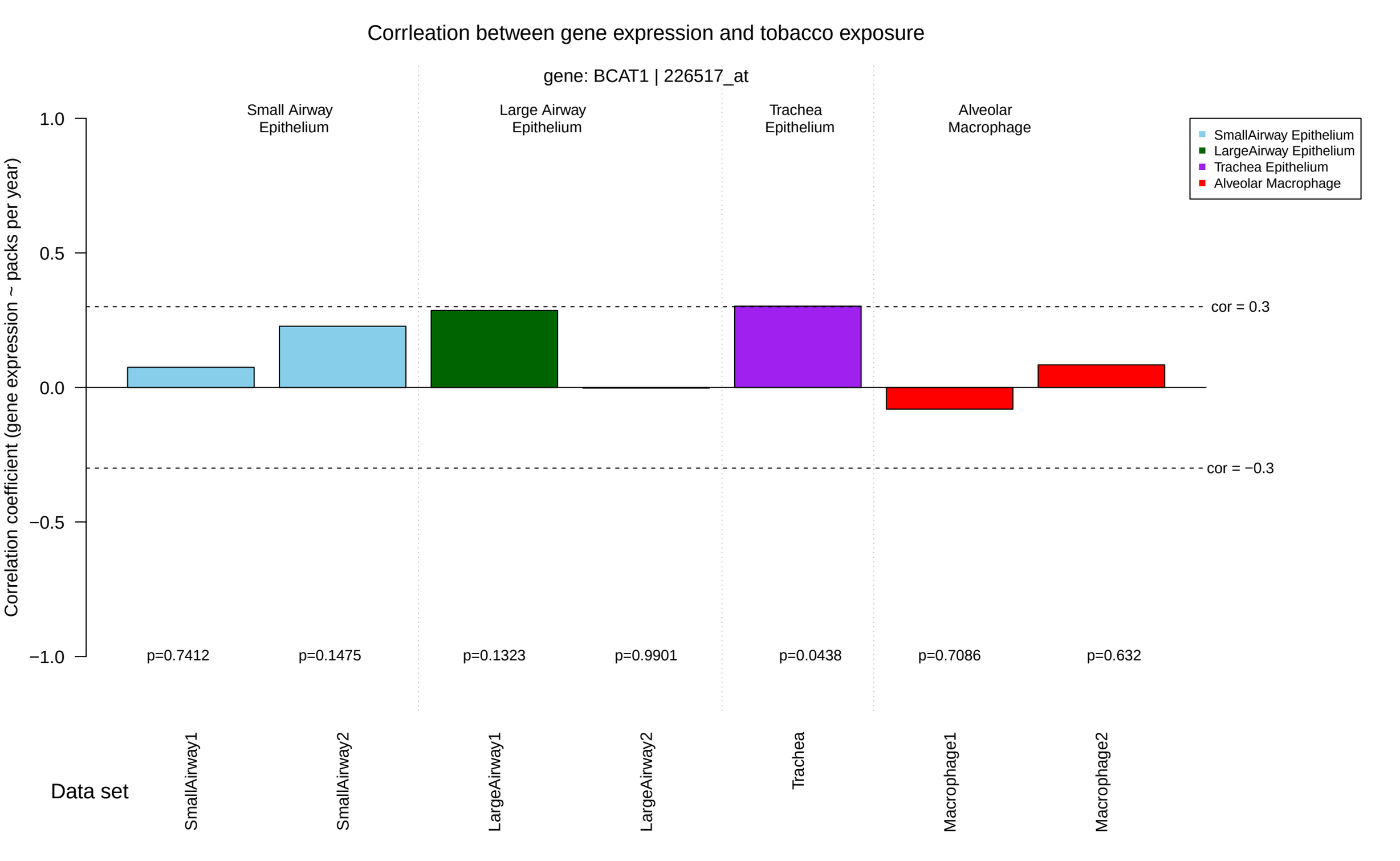

set: LargeAirway1 (p=0.0418)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9358)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0272)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5827)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4594)
BCAT1 | 226517_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BCAT1 | 226517_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7412
p=0.1475
p=0.1323
p=0.9901
p=0.0438
p=0.7086
p=0.632
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2